

AI Trends in 2024

- Exploring the Latest Developments
- Machine Learning to Agent Systems

Machine Learning Fundamentals

- Supervised and unsupervised learning
- Deep learning breakthroughs
- Transfer learning applications

Transformers Revolution

- Attention mechanisms
- Self-supervised learning
- Foundation models at scale

Generative AI

- Text generation models
- Image synthesis and creation
- Multimodal capabilities

Large Language Models (LLMs)

- GPT and similar architectures
- Few-shot learning capabilities
- Prompt engineering techniques

Multimodal AI Systems

- Vision and language integration
- Audio-visual processing
- Cross-modal understanding

AI Agent Systems

- Agent orchestration patterns
- Tool integration and planning
- Multi-agent workflows

Safety and Ethics

- Alignment and safety research
- Bias detection and mitigation
- Responsible AI deployment

Future Outlook

- Emerging applications
- Scalability challenges
- Regulatory landscape

Thank You

- Questions and Discussion